Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for: "Management of Major Haemorrhage"

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SUMMARY

Management of a major haemorrhage requires rapid input from a multidisciplinary team.

Early intervention and effective communication between clinical teams and laboratory staff improves outcome.

Effective communication is vital and improves rapid provision of blood and blood products.

Royal Devon University Healthcare

MAJOR HAEMORRHAGE PROTOCOL RDUH
Give 1g TXA unless GI Bleed

- " I want to activate the Major Haemorrhage Protocol" give location
- 1 person to communicate with: Lab #6151 Eastern, Lab Bleep 045 Northern
- Run samples to Lab (do not use pod at Eastern)
- TEG to direct components and clotting products

EMERGENCY GROUP O RBC

- Immediately available
- x2 Lab fridge, x2 Labour Ward fridge, x2 Theatre fridge
- Immediately available

MH PACK 1

- Resus 2:1 RBC:FFP
- x4 RBC (Group specific 20 min from sample receipt in lab)
- x2 FFP (20 min to thaw in lab)

REASSESS

- Maintain Hb >80g/L
- Maintain Platelets > $50x10^9$/L
- If Fibrinogen <1.5g/L (use 2 pools cryo or 4-6g fibrinogen concentrate)
- Maintain INR<1.5 (PCC if anticoagulated, use TEG to direct other

STILL BLEEDING:

- Resus 1:1 RBC:FFP
- x4 RBC
- x4 FFP
- Order platelets x2 ATD with 2 hr delivery time Northern

Stopped Bleeding? Stand down

MH PACK 3

- x4 RBC
- x2 Cryoprecipitate / 4-6g Fibrinogen concentrate
- x1 Platelets Eastern

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1. INTRODUCTION

1.1 This document gives guidance on the management of adult Major Haemorrhage at the Royal Devon University Healthcare NHS Foundation Trust (RDUH) hereafter referred to as the Trust.

1.2 Its aim is to improve communication between clinical teams and laboratory staff to provide effective resuscitation of patients and rapid provision of blood components and blood products.

1.3 There are separate flow charts for:

- Code Red
- Major Obstetric Haemorrhage (MOH)
- Paediatric Major Haemorrhage
- Major Haemorrhage in Community Hospitals (Nightingale)

1.4 All medical, nursing, laboratory and support staff should know how to find this guidance and know their role. The Major Haemorrhage flow charts should be visibly present in high risk areas, as relevant.

1.5 Anticoagulation should be reversed as soon as possible if a major haemorrhage is suspected. Refer to the Trust Anticoagulation Policy.

2. BACKGROUND

2.1 In 2010 the National Patient Safety Agency (NPSA) issued a rapid response report highlighting patient harm due to delay in the provision of blood components in major haemorrhages. One of the actions required was that each Trust should have a local Major Haemorrhage policy.

2.2 The importance of this was further highlighted in the NICE guidance (NG24) (2015) Blood Transfusion, the British Society for Haematology (BSH) guideline for the haematological management of major haemorrhage (2022), CAS Alert (2022) and the national Serious Hazards of Transfusion (SHOT) reports.

3. DEFINITION OF A MAJOR HAEMORRHAGE

1. Major haemorrhage is arbitrarily defined as:
   - The loss of one blood volume within a 24-hour period
     or
   - Replacement of one blood volume or the transfusion of 10 or more units within a 24-hour period
     or
   - Replacement of 50% blood volume within 3 hours or less
     or
   - Ongoing transfusion requirement in an adult to compensate greater than 150ml / min blood loss
     or
   - Bleeding leading to a heart rate of greater than 110 beats/min and / or a systolic blood pressure considered to have dropped significantly from patient's base line.

3.2 In the Obstetric patient; postpartum haemorrhage is defined as blood loss greater than 1000mls and Major Obstetric Haemorrhage is greater than 1500mls.

3.3 Blood loss is normally underestimated and haemoglobin and haematocrit values may not fall for several hours so do not be falsely reassured by one normal value.

4. ACTIVATING THE MAJOR HAEMORRHAGE ALERT

4.1 Early Communication with the laboratory is crucial to the timely delivery of appropriate blood components.

4.2 Dial **2222** stating "I want to initiate the Major Haemorrhage Protocol" and give your location.

4.3 Switchboard puts the call through to:

Team	Eastern	Northern
Transfusion Laboratory	#6151	Bleep 045
Transfusion Nurse		Bleep 280
Site Management	Bleep 500	Bleep 500
Chargehand Porter		Bleep 506

Anaesthetic on call	Bleep 484	Bleep 508
Anaesthetic 4th on call /consultant	Bleep 489	

4.4 A nominated member of the clinical team to contact the laboratory staff with the patient name, identification number, date of birth and details of the situation. The laboratory staff will be able to inform whether the patient has a valid group and save sample, or if a sample (or two) needs to be taken.

4.5 It is essential that the same member of clinical staff speaks with the laboratory staff at all times so as to optimise communication (#6151 Eastern or bleep 045 Northern).

4.6 In core hours only, Transfusion Laboratory staff to inform the Transfusion Practitioner (Eastern only).

4.7 In cases of major haemorrhage, if the samples are taken via the interosseous route (Eastern only) the laboratory must be informed as these are processed differently.

4.8 Nominate a member of staff to collect blood components and take blood samples to the laboratory. Samples (if required) must be taken directly to the laboratory rather than put in the pod system (Eastern only).

4.9 The laboratory will give an estimate of the time that it will take to issue red cell components as this will depend on sample availability. FFP should be ready for collection in approximately 20 minutes from request.

4.10 Site management will ensure sufficient staff are available in the clinical area to manage the haemorrhage and collect blood components/ products.

5 ORDERING BLOOD COMPONENTS

5.1 The Major Haemorrhage Order Set is available on EPIC to enable rapid requests for blood components by selecting MH Pack 1 and subsequent packs as required.

6 RESUSCITATION

6.1 During resuscitation with a major haemorrhage think A, B, C

- Maintain airway with adequate ventilation with additional oxygen
- Control the source of the haemorrhage and gain venous access with 2 large bore cannula

- Transfuse blood components/products promptly
- Maintain normothermia
- Anticipate a coagulopathy
- Correct calcium levels

6.2 The blood pressure is adequate if the patient is conscious, talking and has a palpable peripheral (e.g. radial) pulse.

6.3 In cases of major trauma, resuscitation should be with blood components/ products rather than crystalloid wherever possible.

6.4 All patients being resuscitated for major haemorrhage are at risk of dilutional coagulopathy.

6.5 The goals during a major haemorrhage is restoration of an adequate blood volume to maintain tissue perfusion and oxygenation:

- a falling heart rate (less than100 beats per min)
- restoration of blood pressure (greater than100 mmHg systolic)
- satisfactory urine output (greater than 30 mls/hour or 0.5 mg/kg/hr)
- a falling serum lactate on blood gas analysis
- Hb 70-90g/L
- Platelets greater than 50 x 10^9/L (greater than 75 x 10^9/L obstetrics)
- fibrinogen greater than 1.5g/L (greater than 2g/L obstetrics)
- INR less than 1.5

7 INVESTIGATIONS

7.1 Always take the following samples to the laboratory as soon as possible:

- Full blood count
- Group and save
- Clotting screen (INR and APTR) with Clauss fibrinogen
- Urea and Electrolyte

7.2 In addition monitor arterial blood gas including serum lactate.

7.3 Thromboelastometry (TEG) if available should be considered to support detection of coagulopathy and allow goal directed replacement.

7.4 Samples for group and save should ideally be taken prior to administration of emergency group O red cells. Inform the laboratory if the patient has received emergency group O red cells in another hospital or in transit to the hospital. Transfusion of emergency group O red cells can interfere with laboratory testing, potentially causing delays.

7.5 Repeat FBC and coagulation screen after every 4 units transfused or every hour, whichever is sooner.

8 TRANEXAMIC ACID (TXA)

8.1 The British Society of Haematology (BSH) (2022) recommend 1g tranexamic acid (TXA) for all cases of major blood loss with the exception of gastrointestinal bleeding due to the increased risk of venous thromboembolic events and seizures.

8.2 NICE guidelines support TXA for all surgical cases where more than 500ml blood loss is expected.

8.3 BSH recommend that an initial dose of TXA (1g intravenously) is given to women with PPH within 3 hours of bleed onset. If bleeding continues after 30 minutes, or it has stopped and restarted within 24 hours of the first TXA dose, a second dose of 1g should be given.

9 BLOOD CELL SALVAGE

9.1 If possible cell salvage should be set up to enable recycling of the patients own blood and reduce the requirement for allogeneic blood.

9.2 250ml washed salvaged red cells can be considered to be equivalent to 1 unit of packed red cells.

9.3 Cell salvaged blood only contains red cells and if large volumes are reinfused, additional plasma components and clotting products will be needed.

9.4 Bacterial contamination of the wound is a relative contraindication to cell salvage.

10 BLOOD COMPONENTS AND PRODUCTS

Provision of Blood: OPTIONS: Urgency versus Time to Issue*

Immediately*	**20 minutes**	**up to 90 minutes****
Emergency O D-negative / O D-positive red cells	ABO / D compatible but antibody screen not complete	Fully Cross-matched

* Fully compatible blood may be provided within minutes if the Transfusion Laboratory has a valid group and save sample and the patient is eligible for electronic issue or remote allocation (unallocated). Contact the Transfusion staff for advice or check on the BloodTrack Enquiry screen.

** Provided the sample shows no evidence of atypical antibodies. If atypical antibodies are present the provision of fully compatible blood can be delayed. In these cases, the Transfusion staff will make contact to discuss time delay and concessionary release (best matched).

1. Immediately available Emergency Group O red cells

10.1.1 In an emergency when there is insufficient time to wait for red cells from the laboratory labelled for the patient, Emergency Group O red cells should be given if transfusion is immediately required.

10.1.2 Patients with child bearing potential (female less than 51 years of age) should be given Group O RhD negative, K negative red cells in order to avoid

development of anti-D and/or anti-K which can lead to haemolytic disease of the newborn in subsequent pregnancy.

10.1.3 Males less than 18 years of age should also be given Group O RhD negative red cells where there is no time to wait for laboratory issued red cells.

10.1.4 Immediately available Emergency Group O red cells are found:

- Laboratory Blood Fridge: 2 units
- Maternity Blood Fridge: 2 units
- Main Theatre Blood Fridge: 2 units

10.1.5 A BloodTrack pick-up slip is NOT required to collect emergency group O red cells from the blood fridges but staff must have BloodTrack access. BloodTrack Tx must be used to administer all units to the patient.

10.2 Group specific or crossmatched red cells

10.2.1 Group specific or crossmatched red cells should be given at the earliest possible opportunity to preserve our stocks of Emergency Group O as this is the most appropriate blood for patients once their blood group is known.

10.3 Electronic Issue (EXM)

10.3.1 Patients with a valid sample may be eligible for Electronic Issue (EXM) which can be provided within minutes.

10.4 Blood components/products for replacement of clotting factors

	Dose	Volume ml	Factor VIII IU/pack	Fibrinogen mg
Fresh frozen plasma (FFP)	15ml/kg	~200	160	520
Cryoprecipitate	x2 packs	~236	499	1596
Fibrinogen Concentrate	4g	100ml	0	4000

10.5 Fresh Frozen Plasma (FFP)

10.5.1 An initial ratio of 2:1 red cells to FFP for resuscitation is suitable for most cases for the first 4 units of red cells, after this a more aggressive approach is taken with a 1:1 ratio.

10.5.2 In obstetrics, because of the increased levels of clotting factors found in pregnant women, red cells are initially used. If further factors are needed, cryoprecipitate is generally used so as not to dilute existing factors.

10.5.3 FFP is given at a dose of 15 ml/kg after the loss of one blood volume (calculated at 70 ml/kg) and definitely given before 1.5 blood volumes have been lost.

10.5.4 Further administration of FFP should be guided by point of care testing (TEG) or coagulation tests aimed at maintaining the INR at less than 1.5.

10.5.5 FFP takes 20 minutes to thaw; infusion should be started within 2 hours of thawing and completed within 4 hours of thawing.

10.6 Cryoprecipitate

10.6.1 This is concentrated frozen source of fibrinogen which also contains factor VIII, factor XIII and von Willebrand factor.

10.6.2 An adult dose is 2 pooled units of cryoprecipitate.

10.6.3 Cryoprecipitate takes 20-30 minutes to thaw; infusion should be started within 2 hours of thawing and completed within 4 hours of thawing.

10.7 Fibrinogen Concentrate

10.7.1 Fibrinogen is the first clotting factor to fall to critical levels during major haemorrhage.

10.7.2 Freeze dried fibrinogen concentrate is made from pooled human plasma.

10.7.4 The standard dose of fibrinogen concentrate is 4g.

10.7.5 It is stored in the transfusion laboratory (Eastern and Northern) and in the anaesthetic room of the emergency obstetric theatre (Eastern only).

10.7.6 BloodTrack Tx must be used to administer the product for traceability purposes.

10.8 Platelets

10.8.1 Platelets should be administered to maintain a platelet count of greater than 50 x 10^9 /L (greater than 75 x 10^9 /L in obstetrics).

10.8.2 Patients who have sustained major trauma or who have an intracranial injury should have a target platelet count of 100 x 10^9/L.

10.8.3 A platelet count of 50 x 10^9/L or less can be anticipated when approximately 2 blood volumes have been replaced; however, individual variation is great.

10.8.4 One adult therapeutic dose (1 ATD) is held as stock for emergencies in Eastern services.

10.8.5 In Northern services, anticipation of platelet requirements should allow for ‘blue light’ delivery time from Plymouth (2 hours 35 minutes).

10.8.6 One adult therapeutic dose (1 ATD) should increase the platelet count by approximately 30 x 10^9/L.

10.8.7 Platelets should be given through a giving set that has not been previously used for red cells as this may reduce the effective transfused platelet dose.

10.9 Recombinant Factor VIIa (rFVIIa)

10.9.1 Recombinant Factor VIIa (rFVIIa) is no longer recommended in the management of major haemorrhage.

11. DISSEMINATED INTRAVASCULAR COAGULOPATHY (DIC)

11.1 DIC is a disorder resulting from inappropriate excessive activation of the haemostatic system. It may be manifested by both thrombotic and haemorrhagic pathology.

11.2 The main trigger for DIC is the exposure of blood to a source of tissue factor (TF) that initiates coagulation, for example from sepsis, tissue injury or malignant cells.

11.3 DIC is more commonly seen in trauma and obstetrics.

11.4 Massive tissue factor exposure results in early fibrin formation leading to activation of clotting factors and subsequent fibrinolysis, this may result in micro thrombus formation.

11.5 There is consumption of nearly all clotting factors and platelets. When combined with fibrinolysis and the generation of fibrin degradation products, this results in the generalised and continued bleeding that is characteristic of DIC.

11.6 Early aggressive replacement is needed focusing specifically on fibrinogen replacement.

11.7 The coagulopathy will worsen with the hypothermia, acidosis and hypocalcaemia that commonly occur with major haemorrhage and subsequent resuscitation.

11.8 Traditional laboratory tests (PT and APTT) do give an indication of coagulability but near patient assays of clot viscoelasticity using TEG, monitor clot strength and fibrinolysis to give a more dynamic picture and can identify impaired platelet function, low fibrinogen levels and low coagulation levels.

12 RISKS OF A MASSIVE BLOOD TRANSFUSION

12.1 Transfusion in the NHS is generally extremely safe, however is not without risk. The Serious Hazards of Transfusion (SHOT) Report in 2024 relates that transfusion delays and pulmonary complications (both TACO and non-TACO) remain leading causes of transfusion-related deaths in the UK accounting together for over 76% of the deaths reported. Notwithstanding that, the

absolute risk of death remains relatively low, at 1 in 58,000 components issued. Harms are at least five times more common.

12.2 One of the greatest risks in major haemorrhage is that the provision of blood components and blood products is delayed.

12.3 Any transfusion related adverse event must be reported via DATIX, reviewed by the Hospital Transfusion Team and reported to SHOT as appropriate.

12.4 Human Error

12.4.1 Human Error, often inpatient identification at blood sampling and administration, is the most common cause for adverse incidents relating to blood transfusion reported to SHOT. Such errors can lead to incompatible ABO blood transfusion causing fatal haemolytic reactions; one of the NHS 'never events'.

12.4.2 It is vital to positively identify patients at every stage of the transfusion process and before administration check blood product compatibility labels against the details on the patient wrist band and prescription as described in the Trust Blood Transfusion policy.

12.4.3 The BloodTrack electronic system is designed to reduce the risk of human error and should be used as the patient safety checks when taking any samples for blood transfusion or administering any blood component/product.

12.5 Transfusion Associated Circulatory Overload (TACO)

12.5.1 TACO is a serious but under-recognised complication of blood transfusion and all patients who are thought to have developed TACO with blood transfusion should be reported via the Hospital Transfusion Team to SHOT.

12.5.2 TACO is defined as patients developing:

- acute or worsening respiratory compromise and/or
- acute or worsening pulmonary oedema during or up to 12 hours after transfusion (SHOT continues to accept cases up to 24 hours) as evidenced by clinical or radiographical findings.
- evidence of fluid overload and cardiovascular system changes not explained by the patient's underlying medical condition

12.5.3 TACO can be associated with rapid or massive transfusion of blood components/products, especially in patients with diminished cardiac function or chronic anaemia. Patients over the age of 60 years of age, infants and those with severe anaemia, are particularly susceptible.

12.6 Hypocalcaemia

12.6.1 High infusion rates of components containing citrate can decrease calcium concentrations, particularly in patients with hypothermia or liver failure (who are unable to metabolise citrate). FFP and platelets contain high citrate concentrations, therefore serum calcium levels must be monitored during massive transfusion.

12.6.2 Calcium chloride should be administered during massive transfusion if ionised calcium levels are low or electrocardiograph (ECG) suggests hypocalcaemia. The recommended dose is 10mls of 10% calcium chloride IV.

12.7 Hyperkalaemia

12.7.1 Hyperkalaemia may occur due to the high extracellular potassium concentration in stored red cell units.

12.7.2 This may be compounded by oliguria associated with shock.

12.7.3 If greater than 6 mmol/L, refer to the Trust guideline for Emergency Management of Hyperkalaemia in Adults.

12.8 Hypomagnesaemia

12.8.1 Hypomagnesaemia is often associated with massively transfused patients and will need monitoring and correction.

12.8.2 For the treatment of hypomagnesaemia (less than 0.6 mmols/L) give 2g (8 mmols Mg2+) over 30 minutes or 5g (20 mmols Mg2+) over 1 hour IV.

12.9 Acid base disturbances

12.9.1 Transfused citrate will produce an alkalosis once it is metabolised.

13 MONITORING COMPLIANCE WITH THIS GUIDELINE

13.1 Monitoring compliance with this policy will be the responsibility of the Hospital Transfusion Team.

13.2 Audits will be reviewed by the Hospital Transfusion Team and presented to the Patient Blood Management Group.

13.3 Where non-compliance is identified, support and advice will be provided to improve practice.

14 ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

- Stanworth SJ, Dowling K, Curry N, Doughty H, Hunt BJ, Fraser L, et al., on behalf of The Transfusion Task Force of the British Society for Haematology. A guideline for the haematological management of major haemorrhage: A British Society for Haematology Guideline. Br J Haematol. 2022;198:654–667. https://doi.org/10.1111/bjh.18275

- National Patient Safety Agency (NPSA) Rapid Response Report (RRR) *The transfusion of blood and blood components in an emergency* NPSA/2010/RRR017

- NICE. Blood transfusion. NICE guideline [NG24]. https://wwwniceorguk/guidance/ng24. November 2015

- RDUH Blood Transfusion Policy

- SHOT Report

- Serious Hazards of Transfusion. Central Alerting System – Preventing Transfusion Delays in Bleeding and Critically Anaemic Patients. 2022. Available at: https://www.cas.mhra.gov.uk/ViewandAcknowledgment/ViewAlert.aspx?AlertID=103190